Shibam **GHOSH**

POSTDOCTORAL RESEARCHER, INRIA, PARIS, FRANCE

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Professional Summary

Postdoctoral Researcher in the COSMIQ team at the French National Institute for Research in Digital Science and Technology (INRIA, Paris), specializing in symmetric-key cryptography, with expertise in cryptanalysis, primitive design, and security evaluation.

Research Interests

- Design and Cryptanalysis of Symmetric-key Primitives
- Security Analysis of Cryptographic Implementations
- Practical Aspects of Cryptanalysis
- Lightweight Cryptography
- Zero-Knowledge Protocols

Education

Indian Statistical Institute — *Kolkata, India*
MASTER OF TECHNOLOGY IN CRYPTOLOGY AND SECURITY — *2018 – 2020*

Presidency University — *Kolkata, India*
MASTER OF SCIENCE IN MATHEMATICS — *2015 – 2017*

Krishnagar Government College — *Krishnagar, India*
BACHELOR OF SCIENCE IN MATHEMATICS — *2012 – 2015*

Research Experience

Institut national de recherche en sciences et technologies du numérique (INRIA) — *Paris, France*
RESEARCH ENGINEER/POSTDOCTORAL RESEARCHER — *Jan 2025 – Present*

- Supervisor: Léo Perrin
- Focus Area: Design and analysis of symmetric-key primitives for advanced cryptographic protocols such as MPC, ZKP, and FHE

Ethereum Foundation — *Remote*
RESEARCH INTERN — *Oct 2024 – Dec 2024*

- Supervisor: Dmitry Khovratovich
- Focus Area: Analysis of post-quantum secure digital signature schemes based on MPC-in-the-head and VOLE-in-the-head

University of Haifa — *Haifa, Israel*
PHD — *Oct 2020 – Dec 2024*

- Supervisor: Prof. Orr Dunkelman
- PhD research focused on symmetric cryptanalysis and primitive design, with key contributions including:
  - Security Analysis of NIST standard AES and SHA3
  - Security Analysis of Arm's Pointer Authentication Code (PAC) FEAT_PACQARMA3
  - Co-designed low-latency tweakable block cipher QARMAv2 for memory encryption
  - Co-designed low-latency tweakable block cipher ChiLow for embedded code encryption
  - Algebraic cryptanalysis of NIST LwC candidates (Ascon, KNOT, TinyJAMBU, Xoodyak)
  - Fault attacks on GIFT and Present
  - Thesis: "Mathematical Techniques in Symmetric-Key Cryptanalysis" (available online)

Institut national de recherche en sciences et technologies du numérique (INRIA) — *Paris, France*
RESEARCH INTERN — *Jan – July, 2020*

- Supervisor: Anne Canteaut, Léo Perrin
- Research focus: Cryptographic Boolean functions and the Big APN Problem
- Developed novel approach using Quadratic Indicator Cubes (QIC) to analyze quadratic APN functions
- Master's Thesis: "On the QIC of quadratic APN functions" (available online)

Publications

Journal Articles

1. Nilanjan Datta, Avijit Dutta, Shibam Ghosh, Eik List, and Hrithik Nandi. HCTR+: an optimally secure tbc-based accordion mode. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2025(3):183–229, 2025, https://doi.org/10.46586/tosc.v2025.i3.183-229

2. Anup Kumar Kundu, Shibam Ghosh, Aikata Aikata, and Dhiman Saha. ToFA: Towards Fault Analysis of GIFT and GIFT-like Ciphers Leveraging Truncated Impossible Differentials. *IACR Trans. Cryptogr. Hardw. Embed. Syst. (TCHES)*, 2025(3):614–643, 2025, https://doi.org/10.46586/tches.v2025.i3.614-643

3. Roberto Avanzi, Orr Dunkelman, and Shibam Ghosh. Differential Cryptanalysis of the Reduced Pointer Authentication Code Function Used in Arm's FEAT_PACQARMA3 Feature. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2025(1):380–419, 2025, https://doi.org/10.46586/tosc.v2025.i1.380-419

4. Marcel Nageler, Shibam Ghosh, Marlene Jüttler, and Maria Eichlseder. AutoDiVer: Automatically Verifying Differential Characteristics and Learning Key Conditions. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2025(1):471–514, 2025, https://doi.org/10.46586/tosc.v2025.i1.471-514

5. Sahiba Suryawanshi, Shibam Ghosh, Dhiman Saha, and Prathamesh Ram. Simple vs. vectorial: exploiting structural symmetry to beat the ZeroSum distinguisher. *Des. Codes Cryptogr. (DCC)*, 93(1):133–174, 2025, https://doi.org/10.1007/s10623-024-01502-x

6. Roberto Avanzi, Subhadeep Banik, Orr Dunkelman, Maria Eichlseder, Shibam Ghosh, Marcel Nageler, and Francesco Regazzoni. The QARMAv2 Family of Tweakable Block Ciphers. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2023(3):25–73, 2023, https://doi.org/10.46586/tosc.v2023.i3.25-73

7. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Practical Related-Key Forgery Attacks on Full-Round TinyJAMBU-192/256. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2023(2):176–188, 2023 , https://doi.org/10.46586/tosc.v2023.i2.176-188

8. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM. *IACR Trans. Symmetric Cryptol. (TOSC)*, 2023(1):41–66, 2023, https://doi.org/10.46586/tosc.v2023.i1.41-66

Conference Papers

9. Avik Chakraborti, Shibam Ghosh, Takanori Isobe, and Sajani Kundu. EWEMrl: A white-box secure cipher with longevity. *IACR Communications in Cryptology*, 2(4), 2026, https://doi.org/10.62056/ak2i5wol7

10. Yanis Belkheyar, Patrick Derbez, Shibam Ghosh, Gregor Leander, Silvia Mella, Léo Perrin, Shahram Rasoolzadeh, Lukas Stennes, Siwei Sun, Gilles Van Assche, and Damian Vizár. Chilow and chichi: New constructions for code encryption. In *Advances in Cryptology – Proceedings of EUROCRYPT*, volume 15601 of *Lecture Notes in Computer Science*, pages 212–243. Springer, 2025, https://doi.org/10.1007/978-3-031-91107-1_8

11. Orr Dunkelman, Shibam Ghosh, Nathan Keller, Gaëtan Leurent, Avichai Marmor, and Victor Mollimard. Partial Sums Meet FFT: Improved Attack on 6-Round AES. In *Advances in Cryptology – Proceedings of EUROCRYPT*, volume 14651 of *Lecture Notes in Computer Science*, pages 128–157. Springer, 2024, https://doi.org/10.1007/978-3-031-58716-0_5

12. Ravi Anand, Shibam Ghosh, Takanori Isobe, and Rentaro Shiba. Quantum Key Recovery Attacks on 4-Round Iterated Even-Mansour with Two Keys. In *Proceedings of Information Security Conference (ISC)*, volume 15257 of *Lecture Notes in Computer Science*, pages 87–103. Springer, 2024, https://doi.org/10.1007/978-3-031-75757-0_5

13. Shibam Ghosh, Anup Kumar Kundu, Mostafizar Rahman, and Dhiman Saha. BIZness: Bit Invariant Zero-Sum Property Based on Division Trail. In *Progress in Cryptology – INDOCRYPT*, volume 15496 of *Lecture Notes in Computer Science*, pages 134–155. Springer, 2024, https://doi.org/10.1007/978-3-031-80311-6_7

14. Anup Kumar Kundu, Shibam Ghosh, Dhiman Saha, and Mostafizar Rahman. Divide and Rule: DiFA - Division Property Based Fault Attacks on PRESENT and GIFT. In *Applied Cryptography and Network Security (ACNS)*, volume 13905 of *Lecture Notes in Computer Science*, pages 89–116. Springer, 2023, https://doi.org/10.1007/978-3-031-33488-7_4

15. Orr Dunkelman, Shibam Ghosh, and Eran Lambooij. Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192 Keyed-Permutation in the Known-Key Setting. In *Progress in Cryptology – INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 349–372. Springer, 2022, https://doi.org/10.1007/978-3-031-22912-1_16

16. Nilanjan Datta, Avijit Dutta, and Shibam Ghosh. INT-RUP Security of SAEB and TinyJAMBU. In *Progress in Cryptology – INDOCRYPT*, volume 13774 of *Lecture Notes in Computer Science*, pages 146–170. Springer, 2022, https://doi.org/10.1007/978-3-031-22912-1_7

17. Shibam Ghosh and Orr Dunkelman. Automatic Search for Bit-Based Division Property. In *Progress in Cryptology – LATINCRYPT*, volume 12912 of *Lecture Notes in Computer Science*, pages 254–274. Springer, 2021, https://doi.org/10.1007/978-3-030-88238-9_13

18. Shibam Ghosh and Léo Perrin. Some Experimental Results on Quadratic APN Functions. In *Boolean Functions and their Applications (BFA) 2021*, 2021 (available online https://boolean.w.uib.no/files/2021/08/BFA_2021_abstract_16.pdf)

EPRINT ARCHIVE

19. Shibam Ghosh, Bastien Michel, and María Naya-Plasencia. Differential-MITM attack on 14-round ARADI. Cryptology ePrint Archive, Paper 2025/1918, 2025

20. Antoine Bak, Shibam Ghosh, Fukang Liu, Willi Meier, Jianqiang Ni, and Léo Perrin. Cryptanalysis of TFHE-friendly cipher FRAST. Cryptology ePrint Archive, Paper 2025/1346, 2025

21. Orr Dunkelman and Shibam Ghosh. Improved key-recovery attacks on ARADI. Cryptology ePrint Archive, Paper 2025/1227, 2025

22. Arghya Bhattacharjee, Ritam Bhaumik, Nilanjan Datta, Avijit Dutta, Shibam Ghosh, and Sougata Mandal. BBB secure arbitrary length tweak TBC from n-bit block ciphers. Cryptology ePrint Archive, Paper 2024/2049, 2024

Research Talks Delivered

2025	**Differential Cryptanalysis of the Arm's FEAT_PACQARMA3 Feature**, FSE	*Rome, Italy*
2025	**Improved key-recovery attacks on ARADI**, Crypto Winter School	*IIT Bhilai, India*
2024	**Partial Sums Meet FFT: Improved Attack on 6-Round AES**, Eurocrypt	*Zurich, Switzerland*
2023	**Attacking the IETF/ISO Standard for Internal Re-keying CTR-ACPKM**, FSE	*Kobe, Japan*
2022	**Full Round Zero-Sum Distinguishers on TinyJAMBU-128 and TinyJAMBU-192**, Indocrypt	*Kolkata, India*
2021	**Automatic Search for Bit-based Division Property**, Latincrypt	*Vitual*
2021	**Some Experimental Results on Quadratic APN Functions**, BFA	*Vitual*

Key Contributions

Enhanced Attack on 6-Round AES *GitHub*

KEY RECOVERY ATTACK **Eurocrypt 2024**

- Fast Fourier Transform (FFT) technique for practical key recovery attacks on 6-round AES
- Full implementation and verification on Amazon AWS cloud infrastructure

Cryptanalysis of ARM Pointer Authentication Code (PAC)

GitHub

SECURITY ANALYSIS — **FSE 2025**

- Cryptanalysis of ARM's **FEAT_PACQARMA3** Pointer Authentication Code (PAC)
- **FEAT_PACQARMA3** utilized in the Arm A-profile and M-profile architectures
- SAT-based automated tool for security assessment against differential attacks
- Co-designed **QARMAv2** block cipher as an alternative to use in Pointer Authentication Code (PAC)

AutoDiVer: #SAT-based Automated Differential Verification

GitHub

CRYPTANALYTIC TOOL — **FSE 2025**

- Automated framework for verifying differential characteristics
- Computing exact probabilities and conditions for the weak-key space using count SAT solvers

ChiLow: Low-Latency Tweakable Block Cipher

PRIMITIVE DESIGN — **Eurocrypt 2025**

- Co-designed tweakable block cipher **ChiLow** for embedded code encryption
- Optimized low-latency primitive securing software code in external memory environments

Advanced Cryptanalysis Techniques in Fault Attacks

GitHub

FAULT ATTACK — **ICHES 2025**

- Advanced cryptanalysis techniques applied to fault attacks on **GIFT** and **GIFT**-like ciphers
- Full implementation and hardware verification using ChipWhisperer Lite platform

Teaching Experience

University of Haifa

Haifa, Israel

TEACHING ASSISTANT — *2022 – 2024*

- Introduction to cryptography, Spring Semester, 2022 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2023 (lecturer in charge: Prof. Orr Dunkelman)
- Introduction to cryptography, Spring Semester, 2024 (lecturer in charge: Prof. Orr Dunkelman)
- Israeli School on Biometrics, 2024 (lecturer in charge: Prof. Orr Dunkelman)

Other Professional Activities

REVIEWER/ PROGRAM COMMITTEE MEMBER

- Journal of Information Security and Applications (JISAS) 2025
- Designs, Codes and Cryptography (DCC) 2025, 2024, 2023, 2022
- Lightweight Cryptography for Security and Privacy (LightSEC) 2025
- Information Security Conference (ISC) 2025
- Finite Fields and Their Applications (FFA) 2025
- (Subreviewer) Eurocrypt 2026, Asiacrypt 2025, Selected Areas in Cryptography (SAC) 2025, 2024

Awards

2021	**Israeli Science Foundation (ISF) Fellowship**, Israeli Science Foundation (ISF)	*Haifa, Israel*
2020	**Data Science Research Center (DSRC) Fellowship**, University of Haifa	*Haifa, Israel*
2020	**Awarded departmental-topper prize**, Indian Statistical Institute	*Kolkata, India*
2019	**Rank 88 in National Eligibility Test**, Council of Scientific and Industrial Research (CSIR), India	*Kolkata, India*

Technical Skills

- C, C++ ●●●●○
- python, SageMath ●●●●○
- Rust ●●●○○
- Git ●●●●○
- Qiskit ●●●○○